

Chapter 11: Logarithmic Functions (Hàm số Logarit)

Chapter Overview

Logarithmic functions are the inverse functions of exponential functions, providing us with powerful tools to solve exponential equations and model phenomena where we need to determine the exponent. From measuring earthquake intensity on the Richter scale to calculating pH levels in chemistry, from determining sound intensity in decibels to analyzing data compression in computer science, logarithms are indispensable in modern science and technology. In this chapter, we explore the definition, properties, and graphs of logarithmic functions, along with their deep connection to exponential functions.

Bilingual Vocabulary (Từ vựng Song ngữ)

Basic Concepts (Khái niệm Cơ bản)

English	IPA	Vietnamese	Example
Logarithm	/ˈlɒɡəˌrɪðəm/	Logarit	The logarithm of 8 to base 2 is 3.
Logarithmic function	/ˌlɒɡəˈrɪðmɪk ˈfʌŋkʃən/	Hàm số logarit	The function $y = \log_2 x$ is a logarithmic function.
Base	/beɪs/	Cơ số	In $\log_3 x$, the base is 3.
Argument	/ˈɑːrgjʊmənt/	Đối số	In $\log_a M$, M is the argument.

Common logarithm	'kɑ:mən 'lɒ:gəɪðəm/	Logarit thập phân	$\log x$ or $\lg x$ denotes the common logarithm (base 10).
Natural logarithm	'nætʃərəl 'lɒ:gəɪðəm/	Logarit tự nhiên	$\ln x$ denotes the natural logarithm (base e).

Properties and Operations (Tính chất và Phép toán)

English	IPA	Vietnamese	Example
Product rule	'prɑ:ɔkt ru:l/	Quy tắc tích	$\log_a(MN) = \log_a M + \log_a N$
Quotient rule	'kwɒʃənt ru:l/	Quy tắc thương	$\log_a\left(\frac{M}{N}\right) = \log_a M - \log_a N$
Power rule	'paʊər ru:l/	Quy tắc lũy thừa	$\log_a(M^k) = k\log_a M$
Change of base formula	/tʃeɪndʒ əv beɪs 'fɔ:rmjələ/	Công thức đổi cơ số	$\log_a M = \frac{\log_b M}{\log_b a}$
Inverse property	'ɪnvɜ:rs 'prɑ:pərti/	Tính chất nghịch đảo	$a^{\log_a M} = M$ and $\log_a(a^x) = x$

Graph and Behavior (Đồ thị và Hành vi)

English	IPA	Vietnamese	Example
---------	-----	------------	---------

Domain	/doʊ'meɪn/	Tập xác định	The domain of $y = \log_a x$ is $(0, +\infty)$.
Range	/reɪndʒ/	Tập giá trị	The range of $y = \log_a x$ is $\mathbb{R}$.
Vertical asymptote	/'vɜ:rtɪkəl 'æsɪmptəʊt/	Tiệm cận đứng	The y-axis is a vertical asymptote of $y = \log_a x$.
Increasing function	/ɪn'kri:siŋ 'fʌŋkʃən/	Hàm đồng biến	When $a > 1$, $y = \log_a x$ is increasing.
Decreasing function	/dɪ'kri:siŋ 'fʌŋkʃən/	Hàm nghịch biến	When $0 < a < 1$, $y = \log_a x$ is decreasing.

Applications (Ứng dụng)

English	IPA	Vietnamese	Example
pH scale	/pi: ɛɪtʃ skeɪl/	Thang pH	pH is calculated using $\text{pH} = -\log[\text{H}^+]$.

Richter scale	<i>/ˈrɪktər skeɪl/</i>	Thang Richter	Earthquake magnitude uses a logarithmic scale.
Decibel	<i>/ˈdesɪbel/</i>	Đềxiben	Sound intensity is measured in decibels using logarithms.
Exponential equation	<i>/ˌɛkspəˈnenʃəl ɪ ˈkweɪʒən/</i>	Phương trình mũ	Logarithms help solve exponential equations.

11.1 Definition of Logarithm (Định nghĩa Logarit)

The Concept of Logarithm

The logarithm answers the question: "To what power must we raise the base to get a certain number?"

Definition: Let

a

be a positive real number with

$a \neq 1$

, and let

M

be a positive real number. The **logarithm of**

M

to base

a

, denoted

$\log_a M$

, is the exponent to which

a

must be raised to produce

M

.

$$\log_a M = x \Leftrightarrow a^x = M$$

Key restrictions:

- The base
 a
must satisfy:
 $a > 0$
and
 $a \neq 1$
- The argument
 M
must be positive:
 $M > 0$
- There is no logarithm of zero or negative numbers

Connection to Exponential Functions

Logarithmic and exponential functions are inverse operations:

- If
 $y = a^x$
, then
 $x = \log_a y$
- If
 $y = \log_a x$
, then
 $x = a^y$

Examples:

- $\log_2 8 = 3$
because
 $2^3 = 8$

- $\log_{10} 100 = 2$
because
 $10^2 = 100$
- $\log_5 \frac{1}{25} = -2$
because
 $5^{-2} = \frac{1}{25}$

Basic Properties from Definition

Fundamental Properties:

For

$$a > 0$$

,

$$a \neq 1$$

,

$$M > 0$$

, and any real number

$$x$$

:

$$1. \log_a 1 = 0$$

(because

$$a^0 = 1$$

)

$$2. \log_a a = 1$$

(because

$$a^1 = a$$

)

$$3. a^{\log_a M} = M$$

(inverse property)

$$4. \log_a(a^x) = x$$

(inverse property)

11.2 Properties of Logarithms (Tính chất của Logarit)

Logarithm Rules

Product Rule:

$$\log_a(MN) = \log_a M + \log_a N$$

The logarithm of a product equals the sum of the logarithms.

Quotient Rule:

$$\log_a\left(\frac{M}{N}\right) = \log_a M - \log_a N$$

The logarithm of a quotient equals the difference of the logarithms.

Power Rule:

$$\log_a(M^k) = k \log_a M$$

The logarithm of a power equals the exponent times the logarithm of the base.

Proof of Product Rule:

Let

$$\log_a M = x$$

and

$$\log_a N = y$$

. Then

$$a^x = M$$

and

$$a^y = N$$

.

Therefore:

$$MN = a^x \cdot a^y = a^{x+y}$$

Taking logarithm base

a

:

$$\log_a(MN) = x + y = \log_a M + \log_a N$$

Change of Base Formula

Change of Base Formula:

$$\log_a M = \frac{\log_b M}{\log_b a}$$

This allows us to convert logarithms from one base to another.

Special cases:

- $\log_a b = \frac{1}{\log_b a}$
(reciprocal relationship)
- $\log_{a^k} M = \frac{1}{k} \log_a M$

Proof:

Let

$$y = \log_a M$$

. Then

$$a^y = M$$

.

Taking

$$\log_b$$

of both sides:

$$\log_b(a^y) = \log_b M$$

Using the power rule:

$$y \log_b a = \log_b M$$

Therefore:

$$y = \frac{\log_b M}{\log_b a}$$

, which means

$$\log_a M = \frac{\log_b M}{\log_b a}$$

11.3 Common and Natural Logarithms (Logarit Thập phân và Tự nhiên)

Common Logarithm (Base 10)

The **common logarithm** is the logarithm with base 10, denoted

$\log x$

or

$\lg x$

.

$$\log x = \log_{10} x$$

Common logarithms are widely used in scientific notation and practical calculations.

Examples:

- $\log 1000 = 3$
(because
 $10^3 = 1000$
)
- $\log 0.01 = -2$
(because
 $10^{-2} = 0.01$
)
- $\log \sqrt{10} = \frac{1}{2}$
(because
 $10^{1/2} = \sqrt{10}$
)

Natural Logarithm (Base e)

The **natural logarithm** is the logarithm with base

$e \approx 2.71828$

, denoted

$\ln x$

.

$$\ln x = \log_e x$$

Natural logarithms are fundamental in calculus and appear naturally in growth and decay models.

Properties of

$\ln x$

:

- $\ln 1 = 0$
- $\ln e = 1$
- $\ln(e^x) = x$
- $e^{\ln x} = x$

Relationship Between Common and Natural Logarithms

Using the change of base formula:

$$\log x = \frac{\ln x}{\ln 10} \approx \frac{\ln x}{2.303}$$

$$\ln x = \frac{\log x}{\log e} \approx 2.303 \log x$$

11.4 The Logarithmic Function (Hàm số Logarit)

Definition

Definition: Let

a

be a positive real number with

$a \neq 1$

. The function

$$f(x) = \log_a x$$

is called a **logarithmic function with base**

a

.

Properties of Logarithmic Functions

For the logarithmic function

$$y = \log_a x$$

where

$$a > 0$$

and

$$a \neq 1$$

:

General Properties:

1. **Domain:**

$$(0, +\infty)$$

(only positive numbers)

2. **Range:**

$$\mathbb{R}$$

(all real numbers)

3. **Continuity:** The function is continuous on its domain

4. **Intercepts:**

- x-intercept:

$$(1, 0)$$

because

$$\log_a 1 = 0$$

- No y-intercept (the graph never crosses the y-axis)

5. **Vertical asymptote:** The y-axis (

$$x = 0$$

) is a vertical asymptote

Monotonicity

Case 1: When

$$a > 1$$

- The function is **increasing** on $(0, +\infty)$

- As
 $x \rightarrow 0^+$
, we have
 $\log_a x \rightarrow -\infty$
- As
 $x \rightarrow +\infty$
, we have
 $\log_a x \rightarrow +\infty$
- For
 $x_1 < x_2$
, we have
 $\log_a x_1 < \log_a x_2$

Case 2: When

$$0 < a < 1$$

- The function is **decreasing** on
 $(0, +\infty)$
- As
 $x \rightarrow 0^+$
, we have
 $\log_a x \rightarrow +\infty$
- As
 $x \rightarrow +\infty$
, we have
 $\log_a x \rightarrow -\infty$
- For
 $x_1 < x_2$
, we have
 $\log_a x_1 > \log_a x_2$

11.5 Graphs of Logarithmic Functions (Đồ thị Hàm số Logarit)

Graph of
 $y = \log_a x$
when
 $a > 1$

The graph has these characteristics:

- Passes through the point
(1, 0)
- Passes through the point
(a, 1)
- Increases from left to right
- Lies entirely to the right of the y-axis
- Approaches the y-axis as
 $x \rightarrow 0^+$
(vertical asymptote)
- Grows slowly as
 $x \rightarrow +\infty$

Table of values for

$$y = \log_2 x$$

:

x	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8
y	-3	-2	-1	0	1	2	3

Graph of
 $y = \log_a x$

when

$$0 < a < 1$$

The graph has these characteristics:

- Passes through the point $(1, 0)$
- Passes through the point $(a, 1)$
- Decreases from left to right
- Lies entirely to the right of the y-axis
- Approaches the y-axis as $x \rightarrow 0^+$
(vertical asymptote)
- Decreases slowly as $x \rightarrow +\infty$

Relationship Between Exponential and Logarithmic Graphs

The graphs of

$$y = a^x$$

and

$$y = \log_a x$$

are **reflections of each other across the line**

$$y = x$$

.

This is because they are inverse functions:

- If (p, q) is on the graph of $y = a^x$, then (q, p) is on the graph of

$$y = \log_a x$$

- The domain of

$$y = a^x$$

equals the range of

$$y = \log_a x$$

- The range of

$$y = a^x$$

equals the domain of

$$y = \log_a x$$

Worked Examples (Ví dụ Mẫu)

Example 1: Evaluating Logarithms

Problem: Calculate the following without a calculator:

a)

$$\log_2 \frac{1}{8}$$

b)

$$\log_3 9$$

c)

$$\log_{\sqrt{3}} 9$$

Solution:

a)

$$\log_2 \frac{1}{8} = \log_2 2^{-3} = -3$$

b)

$$\log_3 9 = \log_3 3^2 = 2$$

c)

$$\log_{\sqrt{3}} 9 = \log_{3^{1/2}} 3^2$$

Using the change of base property:

$$\log_{a^k} M = \frac{1}{k} \log_a M$$

$$\log_{\sqrt{3}} 9 = \frac{1}{1/2} \log_3 9 = 2 \times 2 = 4$$

Alternatively, let

$$x = \log_{\sqrt{3}} 9$$

. Then

$$(\sqrt{3})^x = 9$$

, so

$$(3^{1/2})^x = 3^2$$

, giving

$$3^{x/2} = 3^2$$

, thus

$$x/2 = 2$$

and

$$x = 4$$

.

Example 2: Using Logarithm Properties

Problem: Simplify the following expressions:

a)

$$\log_4 2 + \log_4 32$$

b)

$$\log_2 80 - \log_2 5$$

c)

$$\log_3 27^2$$

Solution:

a)

$$\log_4 2 + \log_4 32 = \log_4 (2 \times 32) = \log_4 64 = \log_4 4^3 = 3$$

b)

$$\log_2 80 - \log_2 5 = \log_2 \frac{80}{5} = \log_2 16 = \log_2 2^4 = 4$$

c)

$$\log_3 27^2 = 2 \log_3 27 = 2 \log_3 3^3 = 2 \times 3 = 6$$

Example 3: Expanding and Condensing Logarithms

Problem:

a) Expand:

$$\log_2 \left(\frac{x^3 \sqrt{y}}{z^2} \right)$$

b) Condense:

$$3 \ln x + \frac{1}{2} \ln y - 2 \ln z$$

Solution:

a)

$$\log_2 \left(\frac{x^3 \sqrt{y}}{z^2} \right) = \log_2 (x^3 \sqrt{y}) - \log_2 z^2$$

$$= \log_2 x^3 + \log_2 \sqrt{y} - \log_2 z^2$$

$$= 3 \log_2 x + \frac{1}{2} \log_2 y - 2 \log_2 z$$

b)

$$3 \ln x + \frac{1}{2} \ln y - 2 \ln z = \ln x^3 + \ln y^{1/2} - \ln z^2$$

$$= \ln(x^3 \cdot \sqrt{y}) - \ln z^2$$

$$= \ln \left(\frac{x^3 \sqrt{y}}{z^2} \right)$$

Example 4: Change of Base Formula

Problem: Calculate

$$\log_4 8$$

without a calculator.

Solution:

Method 1: Using the change of base formula with base 2:

$$\log_4 8 = \frac{\log_2 8}{\log_2 4} = \frac{\log_2 2^3}{\log_2 2^2} = \frac{3}{2}$$

Method 2: Direct approach:

Let

$$x = \log_4 8$$

. Then

$$4^x = 8$$

, so

$$(2^2)^x = 2^3$$

, giving

$$2^{2x} = 2^3$$

.

Therefore

$$2x = 3$$

, so

$$x = \frac{3}{2}$$

.

Example 5: Solving Logarithmic Equations

Problem: Solve for

x

:

a)

$$\log_3 x = 4$$

b)

$$\log_2(x - 1) = 3$$

c)

$$\log_5 x + \log_5(x - 4) = 1$$

Solution:

a)

$$\log_3 x = 4$$

Converting to exponential form:

$$x = 3^4 = 81$$

b)

$$\log_2(x - 1) = 3$$

Converting to exponential form:

$$x - 1 = 2^3 = 8$$

Therefore:

$$x = 9$$

Check:

$$\log_2(9 - 1) = \log_2 8 = 3$$

✓

c)

$$\log_5 x + \log_5(x - 4) = 1$$

Using the product rule:

$$\log_5[x(x - 4)] = 1$$

Converting to exponential form:

$$x(x - 4) = 5^1 = 5$$

Expanding:

$$x^2 - 4x = 5$$

Rearranging:

$$x^2 - 4x - 5 = 0$$

Factoring:

$$(x - 5)(x + 1) = 0$$

Solutions:

$$x = 5$$

or

$$x = -1$$

Check domain: For

$$\log_5 x$$

and

$$\log_5(x - 4)$$

to be defined, we need

$$x > 0$$

and

$$x - 4 > 0$$

, so

$$x > 4$$

.

Therefore,

$$x = -1$$

is extraneous (not in the domain).

Answer:

$$x = 5$$

Example 6: Comparing Logarithmic Values

Problem: Without using a calculator, compare:

a)

$$\log_2 5$$

and

$$\log_2 7$$

b)

$$\log_{0.5} 3$$

and

$$\log_{0.5} 5$$

c)

$$\log_3 5$$

and

$$\log_5 3$$

Solution:

a) Since

$$2 > 1$$

and

$$5 < 7$$

, and the logarithmic function with base > 1

is increasing:

$$\log_2 5 < \log_2 7$$

b) Since

$$0 < 0.5 < 1$$

and

$$3 < 5$$

, and the logarithmic function with base < 1

is decreasing:

$$\log_{0.5} 3 > \log_{0.5} 5$$

c) We need to compare

$$\log_3 5$$

and

$$\log_5 3$$

.

Note that

$$\log_3 5 \cdot \log_5 3 = 1$$

(reciprocal property).

Since

$$3 < 5$$

, we have

$$\log_3 5 > 1$$

(because

$$3^1 = 3 < 5$$

).

Therefore

$$\log_5 3 = \frac{1}{\log_3 5} < 1$$

.

Thus:

$$\log_3 5 > \log_5 3$$

Example 7: pH Scale Application

Problem: The pH of a solution is defined as:

$$\text{pH} = -\log[\text{H}^+]$$

where

$$[\text{H}^+]$$

is the hydrogen ion concentration in moles per liter.

a) Find the pH of a solution with

$$[\text{H}^+] = 0.01$$

mol/L.

b) If a solution has $\text{pH} = 7.4$, find

$$[\text{H}^+]$$

.

Solution:

a)

$$\text{pH} = -\log(0.01) = -\log(10^{-2}) = -(-2) = 2$$

The solution has $\text{pH} = 2$, which is acidic ($\text{pH} < 7$).

b) If $\text{pH} = 7.4$, then:

$$7.4 = -\log[\text{H}^+]$$

$$-7.4 = \log[\text{H}^+]$$

$$[\text{H}^+] = 10^{-7.4}$$

Using a calculator:

$$[\text{H}^+] \approx 3.98 \times 10^{-8}$$

mol/L

This is slightly basic ($\text{pH} > 7$), typical of human blood.

Example 8: Richter Scale Application

Problem: The Richter scale magnitude

M

of an earthquake is given by:

$$M = \log\left(\frac{I}{I_0}\right)$$

where

I

is the intensity of the earthquake and

I_0

is a reference intensity.

a) If an earthquake has intensity 1,000,000 times

I_0

, what is its magnitude?

b) How many times more intense is a magnitude 7 earthquake compared to a magnitude 5 earthquake?

Solution:

a)

$$M = \log\left(\frac{1,000,000 \cdot I_0}{I_0}\right) = \log(1,000,000) = \log(10^6) = 6$$

The earthquake has magnitude 6 on the Richter scale.

b) For magnitude 7:

$$7 = \log\left(\frac{I_7}{I_0}\right)$$

, so

$$I_7 = 10^7 I_0$$

For magnitude 5:

$$5 = \log\left(\frac{I_5}{I_0}\right)$$

, so

$$I_5 = 10^5 I_0$$

The ratio is:

$$\frac{I_7}{I_5} = \frac{10^7 I_0}{10^5 I_0} = 10^2 = 100$$

A magnitude 7 earthquake is **100 times more intense** than a magnitude 5 earthquake.

Applications (Ứng dụng Thực tế)

Application 1: Solving Exponential Equations

Problem: A bank account earns 6% annual interest compounded annually. After how many years will an initial deposit of \$100 million grow to at least \$150 million?

Solution:

Using the compound interest formula:

$$A = P(1 + r)^t$$

We need:

$$100(1.06)^t \geq 150$$

Dividing by 100:

$$(1.06)^t \geq 1.5$$

Taking logarithm of both sides:

$$\log(1.06)^t \geq \log(1.5)$$

$$t \log(1.06) \geq \log(1.5)$$

$$t \geq \frac{\log(1.5)}{\log(1.06)}$$

Using a calculator:

$$t \geq \frac{0.1761}{0.0253} \approx 6.96$$

Since deposits are made annually, we need

$$t = 7$$

years.

After at least **7 years**, the account will reach \$150 million.

Application 2: Sound Intensity in Decibels

Problem: The sound intensity level

L

in decibels (dB) is given by:

$$L = 10 \log \left(\frac{I}{I_0} \right)$$

where

I

is the sound intensity and

$$I_0 = 10^{-12}$$

W/m² is the threshold of hearing.

a) Find the decibel level of a normal conversation with

$$I = 10^{-7}$$

W/m².

b) A rock concert measures 110 dB. Find its intensity.

Solution:

a)

$$L = 10 \log \left(\frac{10^{-7}}{10^{-12}} \right) = 10 \log(10^5) = 10 \times 5 = 50$$

dB

A normal conversation is about 50 decibels.

b) If

$$L = 110$$

:

$$110 = 10 \log \left(\frac{I}{10^{-12}} \right)$$

$$11 = \log \left(\frac{I}{10^{-12}} \right)$$

$$\frac{I}{10^{-12}} = 10^{11}$$

$$I = 10^{11} \times 10^{-12} = 10^{-1} = 0.1$$

W/m²

The rock concert has intensity 0.1 W/m^2 , which is **10 billion times** the threshold of hearing!

Practice Exercises (Bài tập Luyện tập)

Exercise 1: Evaluate without a calculator:

- a)
 $\log_5 125$
- b)
 $\log_2 \frac{1}{16}$
- c)
 $\log_9 3$
- d)
 $\ln e^3$

Exercise 2: Use logarithm properties to simplify:

- a)
 $\log_3 6 + \log_3 \frac{3}{2}$
- b)
 $\log_7 98 - \log_7 2$
- c)
 $2 \log_5 x - \frac{1}{3} \log_5 x^3$

Exercise 3: Solve for

- x
- :
- a)
 $\log_4 x = 3$
- b)
 $\log(x + 2) = 1$
- c)
 $\ln(2x - 1) = 2$

Exercise 4: Use the change of base formula to calculate:

a)

$$\log_3 7$$

(express in terms of common logarithms)

b)

$$\log_8 16$$

Review Exercises (Bài tập Ôn tập)

Exercise 1: Properties and evaluation

Evaluate the following:

a)

$$\log_2 32 + \log_2 \frac{1}{4}$$

b)

$$\log_3 81 - \log_3 27$$

c)

$$\log_5 \sqrt{125}$$

d)

$$\log_6 6^{2x}$$

Exercise 2: Expanding and condensing

a) Expand:

$$\log \left(\frac{x^2 y^3}{z} \right)$$

b) Condense:

$$2 \log_3 x - \log_3 y + \frac{1}{2} \log_3 z$$

Exercise 3: Solving logarithmic equations

Solve for

x

:

a)

$$\log_2 x + \log_2(x - 3) = 2$$

b)

$$\log_3(x + 1) - \log_3(x - 2) = 1$$

c)

$$\ln x + \ln(x - 1) = \ln 6$$

Exercise 4: Graphing

Sketch the graph of

$$y = \log_3 x$$

and identify:

- Domain and range
- x-intercept
- Vertical asymptote
- Whether the function is increasing or decreasing

Exercise 5: Real-world application

The population of bacteria in a culture doubles every 3 hours. If the initial population is 500 bacteria:

a) Write a formula for the population

$$P(t)$$

after

t

hours.

b) How long will it take for the population to reach 10,000 bacteria? (Use logarithms to solve.)

Chapter Summary (Tóm tắt Chương)

Key Concepts

1. Logarithm Definition

$$\log_a M = x \Leftrightarrow a^x = M$$

- The logarithm is the inverse operation of exponentiation
- Domain:
 $M > 0$
 ; Base:
 $a > 0$
 $a \neq 1$

2. Fundamental Properties

- $\log_a 1 = 0$
- $\log_a a = 1$
- $a^{\log_a M} = M$
- $\log_a(a^x) = x$

3. Logarithm Rules

- Product:
 $\log_a(MN) = \log_a M + \log_a N$
- Quotient:
 $\log_a\left(\frac{M}{N}\right) = \log_a M - \log_a N$
- Power:
 $\log_a(M^k) = k \log_a M$
- Change of base:
 $\log_a M = \frac{\log_b M}{\log_b a}$

4. Special Logarithms

- Common logarithm:
 $\log x = \log_{10} x$

- Natural logarithm:
 $\ln x = \log_e x$

5. Logarithmic Function Properties

- Domain:
 $(0, +\infty)$
- Range:
 $\mathbb{R}$
- Vertical asymptote:
 $x = 0$
- x-intercept:
 $(1, 0)$
- Increasing when
 $a > 1$
; decreasing when
 $0 < a < 1$

6. Inverse Relationship

- $y = a^x$
and
 $y = \log_a x$
are inverse functions
- Their graphs are reflections across
 $y = x$

Key Formulas

Formula	Description
$\log_a M = x \Leftrightarrow a^x = M$	Definition of logarithm

$\log_a(MN) = \log_a M + \log_a N$	Product rule
$\log_a\left(\frac{M}{N}\right) = \log_a M - \log_a N$	Quotient rule
$\log_a(M^k) = k\log_a M$	Power rule
$\log_a M = \frac{\log_b M}{\log_b a}$	Change of base formula
$\log_a b \cdot \log_b a = 1$	Reciprocal property
$\text{pH} = -\log[\text{H}^+]$	pH scale
$L = 10\log\left(\frac{I}{I_0}\right)$	Decibel scale

Problem-Solving Strategies

1. **Evaluating logarithms:** Express the argument as a power of the base
2. **Simplifying expressions:** Apply product, quotient, and power rules systematically
3. **Solving logarithmic equations:**
 - Combine logarithms using properties
 - Convert to exponential form
 - Check solutions in the original domain
4. **Using change of base:** Convert to common or natural logarithms for calculator use
5. **Real-world applications:** Identify the logarithmic model and apply appropriate formulas

 **Looking Ahead:** In the next chapter, we will explore **exponential and logarithmic equations and inequalities**, learning systematic techniques to solve more complex problems involving these functions. We'll also see how these tools apply to modeling real-world phenomena in science, economics, and technology.